UNIVERSITY OF HELSINKI

Network Analysis

HOMEWORK

Exercise 3

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Problem 1

Problem 2

Result Fig.1 shows results as running greedy algorithm with different configurations.

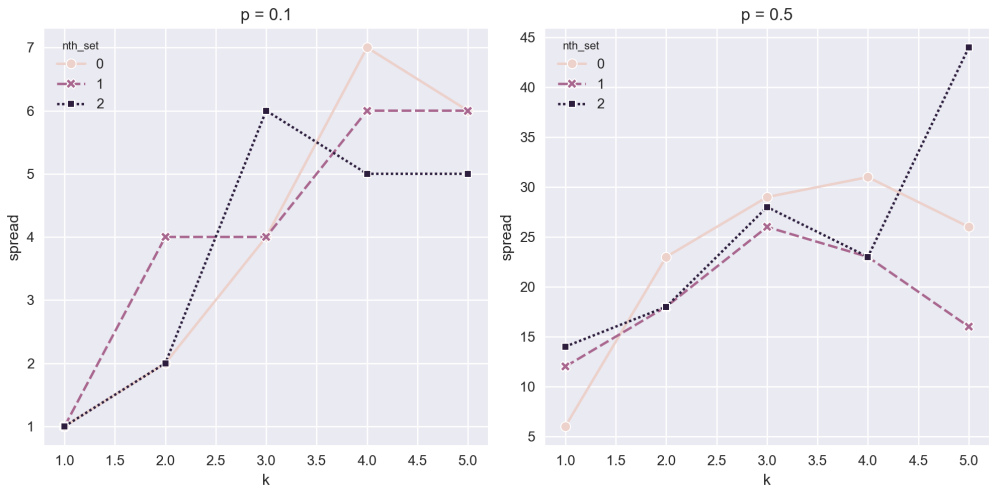

Figure 1: Results of running greedy algorithm with different configurations.

Discussion From Fig.1, it shows that while with infection probability $p = 0.1$ the expected spread shows upward trend, the case of $p = 0.5$ doesn't show consistent trend among different different candidate sets.
It can loosely conclude that the size of initial infected seed doesn't affect the expected spread.

Running time With my personal laptop (Apple M1 Max, 64 GB of RAM), it took about **3 seconds** to run entire code (Fig.2).

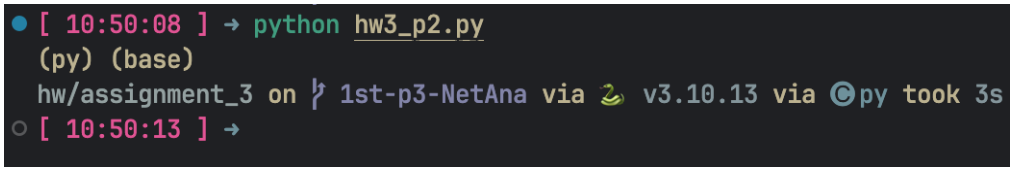

Figure 2: Running time.